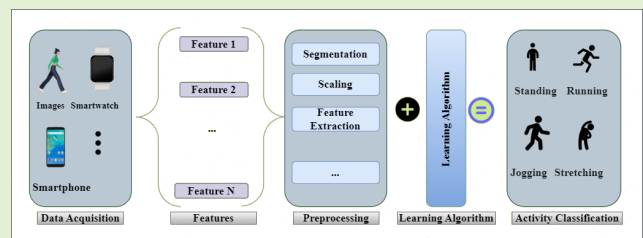


Stack-HAR: Complex Human Activity Recognition With Stacking-Based Ensemble Learning Framework

Abisek Dahal¹, Graduate Student Member, IEEE, Soumen Moulik², Member, IEEE, and Rohan Mukherjee

Abstract—Human activity recognition (HAR) is crucial in various domains, such as healthcare, elderly care, sports, gait analysis, and security surveillance. Despite its critical role in these fields, achieving high accuracy in HAR remains a challenging task. This article proposed a multimodel stacking-based ensemble learning framework for complex HAR (Stack-HAR). The proposed framework demonstrates significant performance improvements across benchmark datasets, such as mHealth, UCI HAR, and WISDM datasets. By leveraging the stacking-based ensemble technique, the proposed Stack-HAR framework achieves exceptional accuracy of 99.49%, 96%, and 90% on these datasets, respectively. The metalearner consistently outperforms individual classifiers, showcasing the effectiveness of ensemble learning. This framework offers a balanced solution for real-world HAR applications, optimizing both accuracy and computational efficiency. Recognizing activities of daily living (ADL) using smartphones, given their widespread sensor availability, gets benefited greatly from this approach.

Index Terms—Activities of daily living (ADL), activity classification, ensemble learning, human activity recognition (HAR), machine learning (ML), sensor fusion, stacking, wearable sensor.



I. INTRODUCTION

IN RECENT years, researchers in artificial intelligence (AI) and human-computer interaction (HCI) have turned their attention to human activity recognition (HAR) because of its wide range of applications. In an era that is dominated by ubiquitous computing, wearable devices, and smartphones, recognizing human activities is crucial for numerous applications related to personal productivity monitoring, health and wellness monitoring, public health and elderly care, security, smart environments, and beyond. For example, tracking of personal activities can significantly enhance efficiency by identifying high-productivity periods and suggesting optimized schedules and workflows. With the world's population aging and digital health becoming a cornerstone of modern healthcare, mon-

itoring daily activities can ensure safety and manage health risks. Human behavior within the realm of ambient-assisted living (AAL), particularly regarding well-being, medical care, and support for the elderly discussed in [1]. HAR involves the identification of various physical activities performed by humans. Various digital devices are used to record the physical activities performed by an individual by capturing data in the form of signals. HAR systems employ machine learning (ML) or deep learning (DL) models, soft computing techniques, and other methods to identify activities by analyzing the signals gathered from wearable devices, environmental sensors, and vision systems [2]. HAR system employing on ML algorithms requires extensive data preprocessing, feature extraction, and many more. With advancements in wearable sensor technology, sensors are becoming smaller and more portable [3]. As a result, HAR systems that utilize wearable sensors have garnered significant interest from researchers across the world.

This article presents some important contributions to the field of HAR those are as follows.

- 1) We propose stacking-based ensemble learning framework for complex HAR (Stack-HAR) a novel multimodel stacking-based ensemble framework for enhanced HAR, which horizontally stacks various models based

Received 28 January 2025; accepted 3 March 2025. Date of publication 17 March 2025; date of current version 1 May 2025. The associate editor coordinating the review of this article and approving it for publication was Prof. Ling Pei. (Corresponding author: Abisek Dahal.)

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Digital Object Identifier 10.1109/JSEN.2025.3549311

on the HAR dataset to create a blended model, utilizing different models as metalearners for optimal activity recognition.

- 2) The proposed framework combines base model predictions with a blended model through metalearning to optimize performance, achieving strong results without the need for complex data preprocessing pipelines and extensive fine-tuning.
- 3) This framework accurately recognizes human activities, thereby enhancing applications in healthcare, fitness monitoring, ambient intelligence, smart homes, elderly care, and personalized health management.

II. RELATED WORKS

HAR algorithms can detect, classify, and understand human activities in real time. A range of studies have explored smartphone-based HAR, highlighting the potential of smartphones for activity recognition in [4], [5], and [6]. The researchers conducted a comparative analysis between support vector machines (SVMs) and alternative classifiers for smartphone data. They use low-pass noise filters for feature selection they attained an accuracy of 72.24% aiming to discern six different activities, namely, running, stair, and climbing [7]. In addition, they use augmented SVM with various different classifiers to enhance accuracy outcomes. Through the combination of SVM with the multilayer perceptron, they achieved a heightened accuracy rate of 91.15% in identifying the same set of six activities. Fu et al. [8] collected data from an IMU and an air pressure sensor at 20 Hz, extracting 19 features. Using joint probability distribution adaptation (JPDA) with improved pseudo labels (IPL-JPDA) method for feature extraction and generalization, they achieved an accuracy of 93.21%. Other models showed an accuracy range of 79%–91%. An independent activity recognition approach that uses orientation-independent features was developed by Siirtola and Rönning [9], comparing it with k -nearest neighbors (kNNs) and quadratic discriminant analysis (QDA). Their models were trained offline using data from eight subjects to enable accurate real-time recognition on smartphones in real-world settings. However, their method does not recognize null data and data that is independent of body position. Also, it does not explore how different types of trousers may impact recognition results. A complex multimodel stacking for HAR, utilizing multilayer stacking and ensemble, is highlighted in [10]. Roitberg et al. [11] introduce a method for HAR in industrial human–robot interaction scenarios. This approach achieved up to 88% accuracy for certain activities, with an average accuracy of 73%. The targeted activities include movement, gestures, and object handling. Image-based HAR in a smart environment was presented in [12]. Their approach utilized background subtraction and skeletization as image processing techniques, coupled with artificial neural network (ANN) for classifying human posture and hidden markov model (HMM) for interpreting activities. A novel method for robust recognition of body pose vector sequences using indexing and sequencing techniques is proposed in [13]. Their approach enables the identification of human activities from videos by analyzing just a few frames and a sequence-based voting

mechanism for accurate recognition. Taha et al. [14] aimed to recognize human activities using 3-D skeleton joints, outperforming some previous methods. They extended existing work by recognizing complex activities as sequences of basic actions using HMM. Their study primarily focused on detecting short-duration actions rather than activities over longer periods. The proposed system may not be suitable for healthcare monitoring due to its reliance on 3-D skeleton information alone, which may limit accuracy compared with additional methods. An introduced an innovative physique-based HAR system that identifies various activities based on physical similarities and ensemble learning is presented in [15]. They discovered that merging data from similar subjects into a single dataset results in more efficient dataset generation and is computationally less demanding than traditional HAR systems. Guo et al. [16] introduced an HAR system incorporating Wi-Fi, IMU, and a hybrid of Wi-Fi + IMU. Their approach utilized Wi-Fi's channel state information (CSI) alongside accelerometer and gyroscope data from IMU devices to capture activity characteristics. Furthermore, Dahal and Moulik [17] proposed a multistream CNN–BiLSTM framework designed to enhance activity recognition by effectively leveraging physiological signals. In recent years, physiological signals have gained significant attention and are increasingly utilized to develop accurate and reliable HAR systems.

Synthesis: Researchers have explored various aspects of ML- and DL-based classification for categorizing the activities performed by individuals or groups. However, these methods are often insufficient for real-world scenarios as they deal with simple activities. Also, the training and evaluation of these models require an extensive amount of data and computational capacities. Most of the activities in studies are conducted in controlled or laboratory-based environments. There is still a lack of effective model fusion for sensor data, as many proposed systems function perfectly only under specific conditions.

III. DATA COLLECTION AND PREPROCESSING

The methodology includes various stages from preparing data to identifying human activities using the proposed framework. Various datasets based on multiple sensors for HAR are accessible online. We have selected three publicly available datasets, namely, UCI-HAR [18], WISDM [19], and mHealth [20] which contain sensor-based activity data. The UCI-HAR dataset consists of 30 volunteers having age between 24 and 49 years, each participating in six activities while wearing a smartphone on their waist. Data are collected using smartphones embedded with three-axial accelerometers and three-axial gyroscopes. In addition, the angular velocity remains constant at a frequency of 50 Hz. In order to process the signals from the sensors, noise filters were used. The readings were then sampled in sliding windows that lasted for a predetermined duration of 2.56 s. These windows had a 50% overlap, meaning that they shared half of their data with the previous window. As a result, each window had 128 readings. The sensor's acceleration signal was divided into two components: body acceleration and gravity. This was achieved by applying a Butterworth low-pass filter.

TABLE I

DETAILED COMPARISON AND DESCRIPTION OF SENSOR-BASED PUBLICLY AVAILABLE BENCHMARK DATASETS

Datasets	# of Activities	Data Collected	# of Subjects	S. Freq.	Sensor Position	Environment
UCI-HAR [18]	6	A, G	30	50Hz	Left Belt, No specific position	Controlled
MHEALTH [20]	12	A, G, M, ECG	10	50Hz	Wrist, Chest, Ankle	Uncontrolled
WISDM [19]	18	A, G	51	20Hz	Smartphone in right pocket	Controlled+Uncontrolled

A = Accelerometer, G = Gyroscope, M = Magnetometer, ECG = Electrocardiogram.

The dataset was assumed to have only low-frequency features with 0.3 Hz as a cutoff frequency. Features were extracted from both time along with the frequency domains for each window, resulting in a comprehensive set of variables for further modeling. The WISDM dataset comprises attributes such as activity, timestamp, and acceleration values across three axes, covering various activities including climbing upstairs and descending downstairs. These six activities are sampled at a rate of 20-Hz translating to one sample every 50 ms, across various users. The mHealth dataset comprises recordings of body motion and vital signs captured from ten volunteers engaging in 12 distinct physical activities. Wearable sensors were affixed to the chest, while the right wrist and left ankle were secured using elastic straps. Each session was recorded at a sampling rate of 50 Hz. In addition, a video was recorded to facilitate detailed activity analysis. This dataset is diverse and can be generalized to common daily activities taking into account variations in body parts used, action intensity, and dynamic execution speed. The activities were conducted in an unconstrained out-of-laboratory environment with participants instructed to exert their best effort during execution. Furthermore, Table I summarizes key details of the sensor-based datasets used, including activities, sensor types, sampling rates, and environments, crucial for assessing the proposed framework's performance. The activity set includes a range of tasks from standing still and sitting to cycling, jogging, and jumping. Minimal preprocessing was applied to the dataset to prepare it for classification, including user data aggregation and metadata-based labeling. We verified the data for completeness, identifying no missing values and few duplicates segmented it into fixed-size windows as per the dataset specifications.

IV. PROPOSED FRAMEWORK

Let us consider m and n as the number of features and samples, respectively, each with an associated activity and corresponding sensor readings from the accelerometer, gyroscope, and other sensors, as shown in Fig. 1. Therefore, the dataset can be represented as follows:

$$\mathbf{X}_{n \times m} = \begin{bmatrix} S_{111} & S_{112} & \cdots & S_{11r_1} & S_{211} & \cdots & S_{21r_2} & \cdots & S_{k1r_k} \\ S_{121} & S_{122} & \cdots & S_{12r_1} & S_{221} & \cdots & S_{22r_2} & \cdots & S_{k2r_k} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ S_{1n1} & S_{1n2} & \cdots & S_{1nr_1} & S_{2n1} & \cdots & S_{2nr_2} & \cdots & S_{knr_k} \end{bmatrix} \quad (1)$$

where $S_{j11}, S_{j12}, \dots, S_{j1r_j}$ are the readings from the j th sensor for the i th sample and k is the number of different sensors. Furthermore, r_j is the number of readings from the j th sensor.

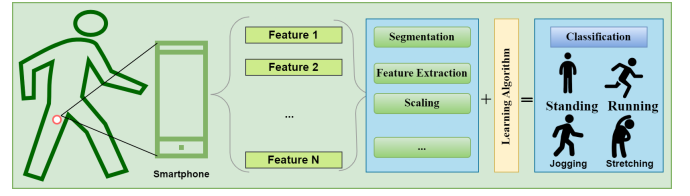


Fig. 1. Proposed generalized system model.

For example, $(A_{x1}, A_{y1}, \text{ and } A_{z1})$, $(G_{x1}, G_{y1}, \text{ and } G_{z1})$, and $(M_{x1}, M_{y1}, \text{ and } M_{z1})$ are the accelerometer, gyroscope, and magnetometer readings, respectively, for the first sample. The corresponding activity labels are given in a vector as follows:

$$\mathbf{y}_{n \times 1} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} \quad (2)$$

where $\mathbf{y}$ is the vector of activity labels corresponding to each row of $\mathbf{X}$. Therefore, $y_i \in \{0, 1, 2, \dots, c\}$, where c is the number of classes.

The dataset is split into 70:30 ratio. Then, training set will consist of $n_{\text{train}} = 0.7 n$ samples, and the testing set will consist of $n_{\text{test}} = 0.3 n$ samples. The training feature matrix $\mathbf{X}_{\text{train}}$ with dimensions $n_{\text{train}} \times m$ and the corresponding training labels vector $\mathbf{y}_{\text{train}}$ with n_{train} elements. Similarly, we construct the validation feature matrix $\mathbf{X}_{\text{test}}$ with dimensions $n_{\text{test}} \times m$ and the corresponding validation labels vector $\mathbf{y}_{\text{test}}$ with n_{test} elements. The training set includes the first 70% of the samples, while the validation set includes the remaining 30%. The preprocessed HAR dataset is provided as input to the base model, while the proposed framework integrates as base models combined to form a blended model. The output of the blended model containing predictions is then fed to the metalearner (the model combines predictions from multiple base models to improve overall performance) which is for efficient feature classification. The proposed framework is mentioned in Fig. 2. The dataset is split in two parts that are represented as $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$ and $(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$.

For the general case, we consider N base models, where N can be a flexible number based on the requirement. Let M_i be the i th base model, where $i \in \{1, 2, \dots, N\}$. Also, each model M_i is trained to optimize its parameters θ_i by minimizing a loss function L over all training instances. For each model M_i , where $i \in \{1, 2, 3, \dots, N\}$, perform κ -fold cross validation. In each fold, training the model on $(\kappa - 1)/\kappa \times n$ samples and test on n/κ samples.

For each base model, M_i is initialize with the model parameters with θ_i is represented as $\theta_i^{(0)} \in \mathbb{R}^p$. The objective for training a model is expressed as follows:

$$\theta_i = \arg \min_{\theta_i} \sum_{j=1}^n L(y_j, f_i(x_j; \theta_i)) \quad (3)$$

where θ_i is the parameters of the i th model. $L(y_j, f_i(x_j; \theta_i))$ is the loss function which measures the changes between the true label y_j and predicted label $f_i(x_j; \theta_i)$. For example, mean squared error (mse) for regression tasks and cross-entropy loss

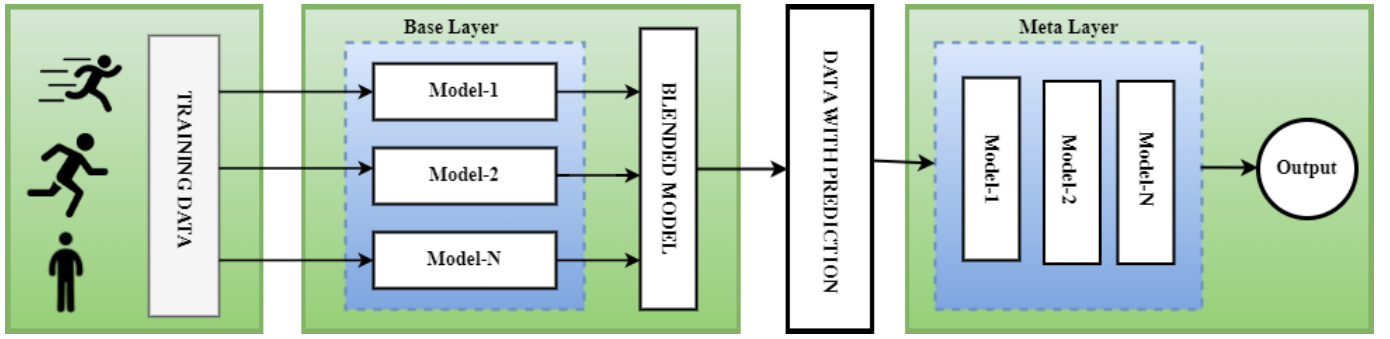


Fig. 2. Proposed Stack-HAR for enhanced human activity recognition.

for classification tasks as activity recognition. j varies from 1 to n since j represents a training instance which varies from 1 to n (sample size is n). f_i is the prediction function of i th model, generating predictions based on the input data x_j and parameters θ_i . The gradient update for each base model is evaluated as follows:

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta \nabla_{\theta_i} L(\theta_i) \quad (4)$$

where η is the learning rate.

Let $P_{i,\text{train}}$ be the class probabilities from the model M_i . We split X_{train} and y_{train} into κ folds. For each fold in $i \in \{1, 2, \dots, \kappa\}$. We train M_i on $(\kappa - 1)/\kappa \times n$ samples and predict class probabilities on the remaining n/κ samples to get P_{i,fold_i} . We combine the predictions of model M_i in the form

$$P_{i,\text{train}} = \begin{bmatrix} P_{i,\text{fold1}} \\ P_{i,\text{fold2}} \\ \vdots \\ P_{i,\text{fold}\kappa} \end{bmatrix}. \quad (5)$$

The predictions P_i generated by the i th base model for the training data are denoted by: $P_i = f_i(X_{\text{train}}; \theta_i)$.

After that, a combine the predictions from the base models is used to form a new dataset $X_{\text{blend},\text{train}}$ for the metalearner. A matrix concatenates the predictions of the N base models, resulting in $c \times n$ features per sample. Therefore,

$$X_{\text{blend},\text{train}} = [P_{1,\text{train}}, P_{2,\text{train}}, \dots, P_{i,\text{train}}]. \quad (6)$$

We initialize metalearner parameters θ_{meta} as $\theta_{\text{meta}}^{(0)} \in \mathbb{R}^q$. The metalearner M_{meta} is trained on the blended dataset $X_{\text{blend},\text{train}}$ and the original labels y_{train} . Therefore,

$$\theta_{\text{meta}} = \arg \min_{\theta_{\text{meta}}} \sum_{j=1}^n L(y_j, f_{\text{meta}}(X_{\text{blend},\text{train},j}; \theta_{\text{meta}})). \quad (7)$$

The trained base models generate predictions on the test data, which are then used to create a blended test dataset $X_{\text{blend},\text{test}}$

$$P_{i,\text{test}} = f_i(X_{\text{test}}; \theta_i). \quad (8)$$

Gradient update for metalearner is calculated as follows:

$$\theta_{\text{meta}}^{(t+1)} = \theta_{\text{meta}}^{(t)} - \eta \nabla_{\theta_{\text{meta}}} L_{\text{meta}}(\theta_{\text{meta}}) \quad (9)$$

$$X_{\text{blend},\text{test}} = [P_{1,\text{test}} \quad P_{2,\text{test}} \quad \dots \quad P_{n,\text{test}}]. \quad (10)$$

The metalearner makes the final predictions using $X_{\text{blend},\text{test}}$

$$\hat{y}_{\text{meta}} = f_{\text{meta}}(X_{\text{blend},\text{test}}; \theta_{\text{meta}}). \quad (11)$$

V. EXPERIMENTAL RESULT

The proposed framework was evaluated using Python within the Jupyter Notebook environment. For training, validation and testing a server equipped with an Intel Xeon Silver 4216 CPU operating at 2.10 GHz was utilized. The server with a multicore processor and is supported by 128 GB of RAM and 1-TB HDD storage. Notably, the model training process did not require GPU usage, as the computational capabilities of the incorporated processor were deemed sufficient for the task at hand. We consider RF, GBC, and k -neighbors as base models with a gradient boosting metalearner that enhances predictive performance by leveraging diverse strengths as it help us to reduce overfitting, balance the bias-variance tradeoff, and improve generalization. Also, these three models performs good in terms of accuracy, precision, and recall as depicted in various tables in results sub section. Gradient boosting as a metalearner further help to refines predictions by correcting errors from base models leading to a more accurate model prediction. To ensure diversity among the selected models, we compute the pairwise correlation of their predictions on a validation set and discard models with correlations.

A. Performance Metric

We evaluated and compared the performance of classification models on standard evaluation metrics, such as precision, recall, $F1$ -score, accuracy, and AUC. Each of these metrics provides unique insights into the model's performance and helps in understanding its effectiveness from different perspectives. Precision (P) evaluates the correctness of the model's positive predictions. It is calculated as the total number of true positives instance (T_p) divided by the total number of predicted positives instance, which includes both true positives and false positives (F_p). Therefore, $P = (T_p)/(T_p + F_p)$. Recall (true positive rate) measures the model potential to identify all actual positive cases. It is determined by the ratio of (T_p) to the total number of actual positives, comprising both true positives and false negatives (F_n). Therefore, $R = (T_p)/(T_p + F_n)$. $F1$ -Score (F_1) represents a balance between precision and recall and a single measure that considers both F_p and F_r . It is the harmonic mean of P and R ,

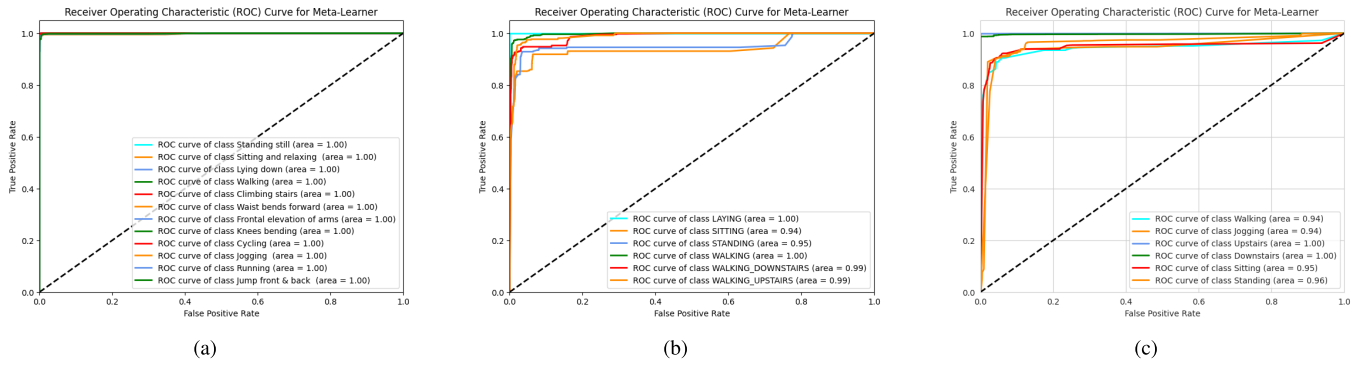


Fig. 3. ROC curves for metalearner performance on (a) mHealth, (b) UCI-HAR, and (c) WISDM datasets.

TABLE II
HYPERPARAMETER INCORPORATED

Classifiers	Hyperparameters
Decision Tree	criterion='gini', splitter='best'
Bagging	base_estimator=None, bootstrap=True
Random Forest	n_estimators=100, criterion='gini'
AdaBoost	base_estimator=None, learning_rate=1.0, algorithm='SAMME.R'
Gradient Boosting	loss='deviance', learning_rate=0.1, subsample=1.0, criterion='friedman_mse'
XGBoost	eval_metric='mlogloss', learn_rate=0.1, max_depth=3
K-Neighbors	n_neighbors=7, weights='distance', algorithm='auto'
Gaussian NB	priors=None, var_smoothing=1e-9
Logistic Regression	penalty='l2', C=1.0, solver='lbfgs'

providing equal importance to both. Therefore, $F_1 = 2 \times (P \times R) / (P + R)$. Accuracy (A) is a simple metric indicating the proportion of correct predictions, including both T_p and true negatives (T_n), out of the total number of predictions. It is calculated as $A = (T_p + T_n) / (T_p + T_n + F_p + F_n)$. Area under the curve (AUC) refers to the area under the receiver operating characteristic (ROC) curve, which plots the T_p rate against the F_p rate at various threshold settings.

B. Results

We evaluated various classifiers and proposed model on three benchmark dataset mHealth, UCI-HAR, and WISDM datasets, assessing their performance across multiple metrics including precision, recall, F_1 score, accuracy, training time, and testing time. The hyperparameters used for each classifier are detailed in Table II. For mHealth dataset from Tables III and IV, the random forest (RF) classifier exhibits the highest performance among the base models, achieving an accuracy of 98.73%, with precision, recall, and F_1 scores all at 0.98. It was used to overcome the limitations of Bagging, especially the issue of correlated trees that limit model diversity. The gradient boosting classifier (GBC) followed closely with an accuracy of 97.99% and precision, recall, and F_1 scores of 0.97. The kNN classifier, despite its simplicity, also performs robustly with an accuracy of 96.86%. However, it is the metalearner, which combines the predictions from these base models, that achieves the highest performance. The proposed framework metalearner reached an accuracy of 99.49%, with precision, recall, and F_1 scores all at 0.99, although at the cost of a significant increase in training time 5922.39 s, as shown in Tables IV–IX, which highlight the performance summary of all models across the datasets, including proposed framework. The activitywise performance of the proposed framework on the mHealth dataset is mentioned in Table III showed perfect

TABLE III
DETAILED ACTIVITYWISE PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR WITH RESPECT TO mHealth DATASET

Activity	P	R	F_1
Standing still	1.00	1.00	1.00
Sitting relaxing	1.00	1.00	1.00
Lying down	1.00	1.00	1.00
Walking	1.00	1.00	1.00
Climbing stairs	0.99	0.99	0.99
Waist bends forward	1.00	1.00	1.00
Frontal elevation of arms	0.99	1.00	1.00
Knees bending	1.00	1.00	1.00
Cycling	0.99	0.99	0.99
Jogging	0.99	0.99	0.99
Running	0.98	0.99	0.99
Jump front back	0.99	0.97	0.98
Accuracy	99.49 %		
M.Avg	0.99	0.99	0.99
W.Avg	0.99	0.99	0.99

scores (precision, recall, and F_1) of 1.00 for activities such as standing still, as well many other activities like waist bends forward. Other activities, such as climbing stairs and jogging, also had high scores, with the lowest performance seen in the “jump front and back” activity, which had an F_1 score of 0.98. The overall high accuracy and detailed activitywise performance underscore the robustness of the proposed framework in UCI-HAR dataset. For the WISDM dataset, RF classifier achieves high performance with an accuracy of 92%, precision, recall, and F_1 scores all at 0.92. The GBC has a slightly lower accuracy of 83.70% but with a notable training time of 1278.98 s. The kNN classifier’s accuracy is 86.16% mentioned in Table VIII. The metalearner outperformed the base models, achieving an accuracy of 90%, with precision, recall, and F_1 scores all at 0.90. In the UCI-HAR dataset, the RF classifier again demonstrated strong performance with an accuracy of 96.70%, followed by gradient boosting with an accuracy of 96.86%. The kNN classifier achieves a respectable accuracy of 91.10% as depicted in Table IX. The proposed framework attained an accuracy of 96.87%. Despite the increased training time, in which proposed framework performance highlights the effectiveness in enhancing activity classification accuracy. The detailed activitywise performance for UCI-HAR is shown in Table X which reflect the perfect scores for “laying,” and high scores for other activities. The overall accuracy of 96% confirms the effectiveness of the proposed framework human activities classification. The activitywise performance

TABLE IV
PERFORMANCE SUMMARY FOR mHealth DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
Decision Tree	0.95	0.95	0.95	95.90	15.55	0.05
Bagging	0.95	0.95	0.96	95.38	95.16	0.48
Random Forest	0.98	0.98	0.98	98.73	168.15	4.13
AdaBoost	0.38	0.36	0.31	38.00	55.52	2.13
Gradient Boosting	0.97	0.97	0.97	97.99	3061.26	3.09
XGBoost	0.97	0.97	0.97	97.77	50.93	4.50
K-Neighbors	0.96	0.96	0.96	96.08	0.07	142.43
Gaussian NB	0.82	0.81	0.81	84.10	0.21	0.19
Logistic Regression	0.70	0.69	0.68	72.95	84.41	0.70

TABLE V
PERFORMANCE SUMMARY FOR UCI-HAR DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
Decision Tree	0.95	0.95	0.95	95.71	5.40	0.05
Bagging	0.96	0.96	0.96	96.43	7.28	0.07
Random Forest	0.96	0.96	0.96	96.70	15.48	0.06
AdaBoost	0.95	0.95	0.95	93.10	33.75	0.21
Gradient Boosting	0.96	0.96	0.96	96.86	1026.35	0.11
XGBoost	0.94	0.93	0.93	93.93	36.49	0.10
K-Neighbors	0.90	0.89	0.89	90.02	15.05	1.05
Gaussian NB	0.79	0.76	0.76	77.03	0.09	0.07
Logistic Regression	0.96	0.96	0.96.13	96.13	48.94	0.02

TABLE VI
PERFORMANCE SUMMARY FOR WISDM DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
Decision Tree	0.90	0.90	0.90	90.00	6.86	0.13
Bagging	0.92	0.92	0.92	92.00	54.35	1.24
Random Forest	0.92	0.92	0.92	92.00	349.19	15.82
AdaBoost	0.33	0.41	0.35	50.00	43.13	3.60
Gradient Boosting	0.78	0.75	0.77	78.00	998.58	3.23
XGBoost	0.85	0.84	0.85	85.00	47.76	3.62
K-Neighbors	0.67	0.62	0.64	67.00	1.12	19.98
Gaussian NB	.08	0.17	0.10	39.00	0.18	0.06
Logistic Regression	0.07	0.17	0.09	39.00	5.28	0.03

TABLE VII
DETAILED PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR FRAMEWORK WITH mHealth DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
K-Neighbours (Base 1)	0.96	0.96	0.96	96.86	15.55	0.05
Random Forest (Base 2)	0.98	0.98	0.98	98.73	168.15	4.13
Gradient Boosting (Base 3)	0.98	0.98	0.98	98.19	3061.26	3.09
Meta-Learner (Meta 1)	0.99	0.99	0.99	99.49	5922.39	10.14

TABLE VIII
DETAILED PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR FRAMEWORK WITH WISDM DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
K-Neighbors (Base 1)	0.89	0.89	0.89	86.16	8.32	0.12
Random Forest (Base 2)	0.92	0.92	0.92	92.18	325.81	0.14
Gradient Boosting (Base 3)	0.83	0.83	0.83	83.70	1278.98	0.09
Meta-Learner (Meta 1)	0.91	0.91	0.91	91.00	1399.69	0.5

TABLE IX
DETAILED PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR FRAMEWORK WITH UCI-HAR DATASET

Model	P	R	F_1	Accuracy (k=5) (%)	Training Time (s)	Testing Time (s)
K-Neighbours(Base 1)	0.91	0.91	0.91	91.10	19.40	0.08
Random Forest(Base 2)	0.96	0.96	0.96	96.70	15.48	0.06
Gradient Boosting (Base 3)	0.96	0.96	0.96	96.86	1026.35	0.11
Meta-Learner(Meta 1)	0.96	0.96	0.96	96.87	1399.69	5.00

on the WISDM dataset in Table XI shows high scores for “jogging,” “upstairs,” “sitting,” and “downstairs,” with slightly lower scores for “walking” and “standing.” The overall

accuracy of 90% and detailed activitywise results indicates that the proposed framework is effective in distinguishing between different physical activities. The proposed framework which

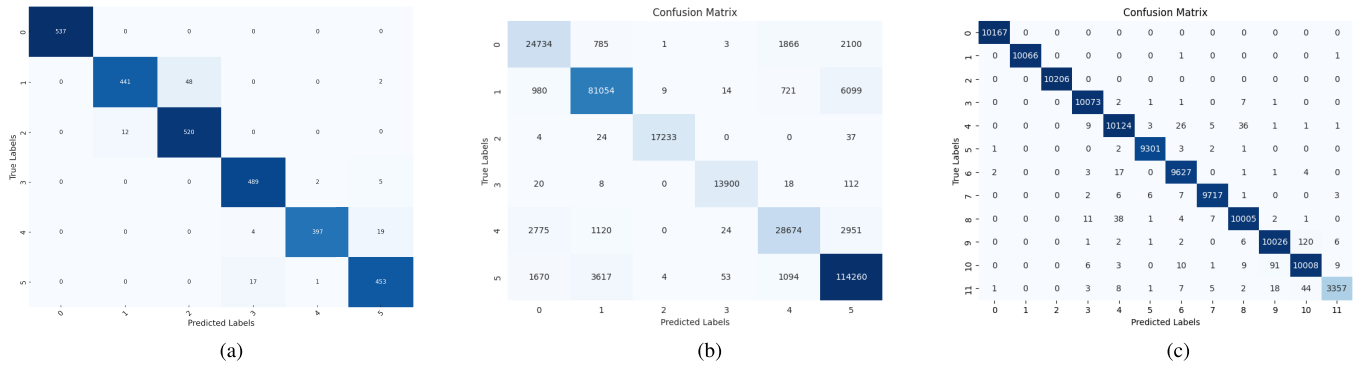


Fig. 4. Confusion matrices of the performance of our proposed Stack-HAR framework on benchmark public datasets. (a) UCI-HAR dataset. (b) WISDM dataset. (c) mHealth dataset.

TABLE X

DETAILED ACTIVITYWISE PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR WITH RESPECT TO UCI-HAR DATASET

Activity	P	R	F_1
Laying	1.00	1.00	1.00
Sitting	0.97	0.90	0.93
Standing	0.92	0.98	0.95
Walking	0.96	0.99	0.97
Walking Downstairs	0.99	0.95	0.97
Walking Upstairs	0.95	0.96	0.95
Accuracy	96%		
M.Avg	0.96	0.96	0.96
W.Avg	0.96	0.96	0.96

TABLE XI

DETAILED ACTIVITYWISE PERFORMANCE COMPARISON OF THE PROPOSED Stack-HAR WITH RESPECT TO WISDM DATASET

Activity	P	R	F_1
Walking	0.82	0.84	0.83
Jogging	0.94	0.91	0.92
Upstairs	1	1	1
Sitting	0.99	0.99.00	0.99
Standing	0.89	0.81	0.84
Downstairs	0.91	0.95	0.93
Accuracy	91.47 %		
M.Avg	0.92	0.91	0.92
W.Avg	0.91	0.91	0.91

uses the multimodel stacking and ensemble approach, achieves higher accuracy lies in its ability to leverage the strengths of multiple base models in our case we have used three base models. Each base model captures different aspects of the data, and by combining their predictions, the metalearner can make more accurate and robust predictions. The proposed approach reduces the risk of overfitting and enhances generalization, leading to improved performance across diverse activities and diverse datasets. The confusion matrix illustrates the high accuracy of the proposed framework in correctly classifying activities, while the ROC curve in Fig. 3(a) for mHealth dataset is close to 1. The ROC curve in Fig. 3(b) shows that most of the activities with an area under the curve close to 1. In addition, Fig. 3(c) demonstrates its excellent discrimination capability, with an area under the curve close to 1. Furthermore, the confusion matrix of the proposed framework, as shown in Fig. 4, provide a clear visualization of its performance. The accuracy of performance of our proposed framework on different datasets is mentioned in Fig. 5.

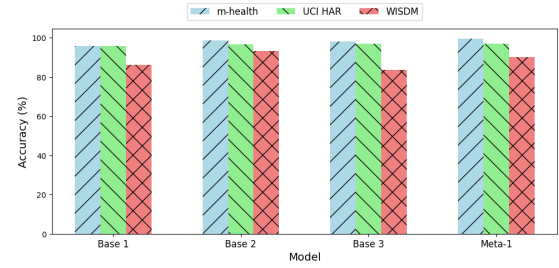


Fig. 5. Accuracy of the proposed Stack-HAR framework.

TABLE XII

SUMMARY OF PERFORMANCE COMPARISON OF Stack-HAR ACROSS DATASETS

Datasets	Acc.(%)	P	R	F1	T.T(s)	Tes.T(s)
mHealth	99.49	0.99	0.99	0.99	5922.39	10.14
UCI-HAR	96.87	0.96	0.96	0.96	1399.69	5.00
WISDM	91.47	0.91	0.91	0.92	1399.69	0.50

Accuracy (Acc.), Precision (P), Recall (R), F1-Score (F1), Training Time (T.T), and Testing Time (Tes.T)

Moreover, WISDM dataset presents unique challenges due to its reliance on accelerometer data only. Unlike multisensor datasets, such as UCI-HAR and mHealth, WISDM limits the granularity of motion information, particularly for dynamic activities like “walking.” While the Stack-HAR framework demonstrates robust performance across datasets, its slightly lower improvement over RF on WISDM can be attributed to the limited feature diversity.

In addition to Table XII presents a performance comparison of the proposed Stack-HAR framework across three benchmark datasets: mHealth, UCI-HAR, and WISDM. The framework achieved the highest accuracy of 99.49% on mHealth and 96.87% on UCI-HAR, outperforming base models in all metrics. However, on the WISDM dataset, it achieved a slightly lower accuracy of 90%, primarily due to the limitations of using accelerometer data alone, which reduces the model’s ability to capture dynamic activities like ‘Walking.’

VI. COMPLEXITY ANALYSIS

The complexity analysis of the proposed framework begins based on computational demands at each step. The training

of the base models involves significant complexity. kNN classifier, training complexity is $O(1)$, as kNN does not involve an actual training phase but rather stores the training data. For RFC, with T trees (in this case, $T = 100$), the training complexity remains $O(100 \cdot m \cdot n \cdot \log(n))$. Similarly, GBC, also with $T = 100$, shares the same training complexity of $O(100 \cdot m \cdot n \cdot \log(n))$. Generating predictions for each model has different complexities. For kNN, the prediction complexity is $O(n \cdot m)$ per query, leading to a total prediction complexity of $O(n^2 \cdot m)$ for all samples. For the kNN classifier, RF, and gradient boosting models, the prediction complexity is $O(m \cdot n)$. Combining predictions from these models into a blended dataset and performing concatenation operations contribute a complexity of $O(n)$ per model, leading to an overall complexity of $O(3n) \approx O(n)$ for the three models. Training the metalearner, here using the GBC, on the blended dataset involves a complexity of $O(100 \cdot k \cdot n \cdot \log(n))$, where k represents the number of features obtained from the base models' outputs.

The complexity associated with predictions of the metalearner on test data is $O(k \cdot n)$, which matches the prediction complexity of the base models. Combining these predictions for final evaluation remains $O(n)$. The final steps include the calculation of accuracy and generation of a classification report, both with a complexity of $O(n)$. The complexity is predominantly influenced by the training phases of the RF and gradient boosting models, both as base models and as metalearner. The total complexity can be approximated as $O(100 \cdot m \cdot n \cdot \log(n)) + O(100 \cdot k \cdot n \cdot \log(n))$. Given $k \approx 3$ m, this simplifies to $O(400 \cdot m \cdot n \cdot \log(n)) \approx O(m \cdot n \cdot \log(n))$.

VII. CONCLUSION AND FUTURE WORK

The proposed Stack-HAR framework demonstrates significant improvement in classifying complex improvement in classifying complex across multiple datasets. By leveraging the strengths of various base models through stacking and ensemble techniques, the framework achieves higher accuracy and robustness in recognizing diverse human activities. The proposed framework achieves remarkable accuracy of 99.49% in mHealth dataset, 96% in UCI HAR, and 90% in WISDM dataset. The metalearner, which integrates predictions from the base models, consistently outperforms individual classifiers, highlighting the effectiveness of ensemble learning methods. This proposed framework enhances predictive accuracy and provides a comprehensive solution for real-world applications in HAR, balancing the tradeoff between accuracy and computational efficiency. The results underscore the potential of Stack-HAR as a powerful framework for advancing the capabilities of activity recognition systems. In future, we will focus on optimizing the computational efficiency of the Stack-HAR

framework and exploring its applicability to real-time HAR scenarios.

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